

Histograms of Oriented Gradients for Human Detection

Navneet Dalal & Bill Triggs — INRIA Rhone-Alps, France

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1. What is This Paper About?

Detecting people in images is not simple. People wear different clothes, stand in different ways, and appear in front of all kinds of backgrounds. This paper asks one key question: what is the best way to describe a human body in an image so that a computer can reliably spot it?

The authors came up with a method called HOG (Histogram of Oriented Gradients). The idea is: instead of looking at raw pixel colours, look at the edges in the image and record which direction they point. A person's outline creates a very recognisable edge pattern, and HOG captures it well.

The three main goals were:

- Prove that HOG works better than older methods like Haar wavelets and SIFT.
- Find out exactly which design choices inside HOG matter the most.
- Introduce a harder test dataset — the INRIA pedestrian dataset with over 1,800 images — since the older MIT dataset was already too easy.

2. How Does the Method Work?

The HOG detection pipeline works in four simple steps:

Step 1 — Find the Edges in the Image

- A simple filter is used to find places in the image where brightness changes sharply — these are the edges.
- No blurring is used before this step. Keeping edges sharp is very important.
- For colour images, gradients are checked for each colour channel and the strongest one is kept.

Step 2 — Build a Histogram for Each Small Region

- The image is split into small areas called cells — each one is about 8x8 pixels.
- Inside each cell, edge directions are sorted into 9 groups (called bins) covering 0 to 180 degrees.
- This creates a small bar chart of edge directions for every cell — that is the histogram.

Step 3 — Group Cells Together and Adjust for Lighting

- Nearby cells are grouped into slightly larger overlapping blocks (like a 2x2 group of cells).
- The histograms inside each block are adjusted together so that lighting differences do not affect the result.
- Because blocks overlap, each cell gets adjusted multiple times — this actually improves accuracy.

Step 4 — Use a Classifier to Decide: Person or Not?

- All the adjusted histograms from across the detection window are joined into one long list of numbers.
- This list is fed into a Support Vector Machine (SVM) which was trained to say: person or not a person?
- The detection window slides across the image at different sizes to find people at different distances.

3. Findings ★

This is the most important part of the paper. The authors tested every single design choice carefully to see what really works. Here is what they found:

Finding 1 — HOG is Far Better Than Every Other Method

- On the MIT dataset, HOG got near-perfect accuracy — almost no mistakes at all.
- On the harder INRIA dataset, HOG had more than 10 times fewer wrong detections than the best wavelet method.
- Other methods like PCA-SIFT and Shape Contexts also did much worse than HOG.
- This result clearly showed HOG was the best approach for finding people in images at that time.

Finding 2 — Blurring the Image Before Edge Detection is a Mistake

- This was one of the most surprising discoveries in the paper.
- Even a small amount of blurring dropped the accuracy from 89% down to 80%.
- The reason is that sharp edges carry a lot of information about the shape of a person. Blurring destroys that.
- So the simplest approach — no blurring at all — turned out to be the best one.

Finding 3 — 9 Direction Groups (Bins) is the Best Number

- Using more direction bins improved accuracy — but only up to a point.
- 9 bins covering 0 to 180 degrees gave the best results. Going above 9 bins made almost no difference.
- Using 0 to 360 degrees (full direction range) actually made things worse — probably because clothing colours and busy backgrounds made the exact direction of contrast unreliable.

Finding 4 — Overlapping Blocks are Really Important

- Without any normalisation at all, accuracy dropped by 27%. So this step is not optional — it is essential.
- Using overlapping blocks (where each cell is adjusted multiple times from different angles) improved accuracy from 84% to 89%.
- It looks like extra work, but it genuinely helps because each small region gets looked at from multiple perspectives.

Finding 5 — The Detector Looks at the Outline, Not the Clothing

- By looking at the SVM weights visually, the authors made a really interesting discovery.
- The detector mostly pays attention to the edges of the person's outline — the head, shoulders, and feet — against the background.
- It does not focus on patterns inside the clothing or on the body itself.
- This is exactly why the method works even when people are wearing very different clothes or standing in front of cluttered backgrounds.

- Also, strong vertical edges inside the person's body were found to be a negative sign — they actually tell the detector 'probably not a person', which helps reduce false detections from things like poles or walls.

Finding 6 — Some Other Useful Details

- **Cell size:** 6 to 8 pixel wide cells work best — which is about the same width as a human arm or leg in the training images.
- **Window border:** Keeping 16 pixels of background around the person improved accuracy by 4%. The background around the person actually gives the detector useful context.
- **Colour helps a little:** Using colour information gave a small 1.5% improvement over using grayscale alone.
- **Fancier classifier:** A Gaussian kernel SVM improved accuracy by 3%, but it was much too slow for real-time use.

4. Suggestions for Future Work ★

HOG was a big leap forward, but it still had clear limitations. Below are suggestions for how this work can be taken further — some from the authors, and some additional ideas.

Suggestions from the Authors

- **Make it faster:** Right now the detector checks every single position in the image, which is slow for large photos or videos. A smarter version would quickly skip areas that clearly have no person and only look carefully at the uncertain spots.
- **Use motion information in videos:** This paper only works on still images. For videos, combining HOG with motion information (like how pixels move between frames) would help detect people who are moving fast or half-hidden behind objects.
- **Handle different body poses:** The fixed detection window works well for someone standing upright but not for someone crouching or bending. A model that detects each body part on its own — head, arms, legs — and then combines them would handle a much wider range of positions.

Additional Suggestions

- **Let the model learn the features on its own:** HOG features were designed by hand based on gut feeling about what matters. Deep learning models like CNNs learn

their own features directly from thousands of images and now clearly outperform HOG. Comparing what CNNs learn with what HOG does by hand could give us useful new insights.

- **Crowded places with overlapping people:** When many people stand close together they overlap, and the fixed window approach cannot separate them well. A future system should be able to detect multiple people at the same time, even when they are partly blocking each other.
- **Detecting people in all kinds of poses:** The INRIA dataset mostly has people standing upright. People who are sitting, lying down, or seen from above are barely covered. A better dataset and detector should be able to handle the full range of human postures.
- **Make it fair for everyone:** The training data was collected in France and lacks variety in terms of skin tone, body shape, and clothing style. A detector trained on such limited data may simply not work as well for people who look different from the training images. Future work should use more diverse datasets and check whether the detector works equally well for people from different backgrounds.
- **Add depth information:** Modern cameras can also measure how far away things are. Adding this depth information to HOG would make detection more reliable in dim lighting or when someone's clothing blends into the background.

Conclusion

The HOG paper is one of the most important papers in computer vision. What made it stand out was not just the idea itself, but the careful way the authors tested everything to find what truly works. Their key findings — like avoiding blurring, using 9 orientation bins, and using overlapping blocks — gave the field very clear and practical guidance. HOG stayed as the go-to approach for human detection for many years and is still an important reference today, even as deep learning has taken over most of these tasks.

Reference

Dalal, N., & Triggs, B. (2005). Histograms of oriented gradients for human detection. In *Proceedings of the IEEE CVPR'05*, Vol. 1, pp. 886-893. IEEE.
<https://doi.org/10.1109/CVPR.2005.177>